

Fluger: Streamlining pharmacy booking and patient management with cloud

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[About Fluger](#)

With Google Cloud, Fluger has implemented a scalable, fast and secure cloud infrastructure that supports expansion of its new pharmacy booking system into markets such as the United States.



About Fluger

Founded in 2020 and headquartered in Victoria, Australia, Fluger provides health services management software to market segments such as pharmacy, local government and large corporations. Some features of this platform include digital consent forms, a clinical governance system, a customisable booking system, a live marketing system and appointment reminders.

Industries: Healthcare, Life Sciences

Google Cloud results

- Scales from 10 to 15 users to several thousand in seconds with no negative impact on user experience
- Reduces cloud costs by up to 15 percent
- Enables the business to execute plans for addressing the United States market
- Halves web page response times

Location: Australia

Achieved
100% ϵ
to deliver
quality
end-user
experience

(<https://cloud.google.com/compute/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/kubernetes-engine/>),

Cloud DNS (<https://cloud.google.com/dns/>)

(<https://marketingplatform.google.com/about>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Marketing Platform

(<https://marketingplatform.google.com/about>)

The coronavirus pandemic placed Australia's healthcare system under immense pressure in the early 2020s. Hospitals managed thousands of admissions per day as infectious variants spread and by July 2023, nearly seven million vaccine doses had been administered. This burden, as well as the widely reported burnout experienced by medical and health professionals, demonstrated the need for technology-enabled innovations to streamline processes, reduce workloads and improve care.

Founded in 2020 to develop and market software that enables providers to book influenza vaccination programs for corporate clients, Victorian startup Fluger (<https://fluger.au/>) saw demand for its products and expertise explode as the pandemic took place. Since its inception, the business has managed about 1.5 million vaccinations for corporate health clients. It has expanded from offering booking software to a portfolio of workplace health management tools.

Maintaining a high quality experience during demand peaks

Fluger's core product enables vaccination providers that service corporate clients to coordinate the schedules of health professionals. This allows clients to administer vaccines or health assessments at their premises, and manage vaccine logistics. At the same time, Fluger can ensure effective cold chain management, which monitors vaccines at a safe temperature of between two and eight degrees celsius during transport and storage.

The business's rapid growth is due largely to an increase in delivering pandemic vaccination programs to aged care homes. The success of these contracts prompted the IT team to evaluate cloud providers that could support expansion into new products and markets, while providing the service and experience demanded by clients and end users.

"We needed a cloud service that could accommodate fluctuations in demand for our corporate vaccination booking system without incurring system outages or slow response times during peaks," explains Hamish Mackellar, Founder, Fluger. "For example, a vaccination provider may send a booking link to 10,000 plus team members at a large corporation at, say, 9:10am on a Monday morning, meaning the number of users of our system explodes from 10 to 15, to several thousand in seconds."

Fluger team members conducted a thorough evaluation of multinational cloud providers, looking towards Google Cloud to drive the development of its pharmacy booking system. The speed and reliability of Compute Engine

(<https://cloud.google.com/compute>), combined with personalized support and technical advice about critical issues such as developing effective workflows, prompted the business to elect to move its primary corporate vaccination system to Google Cloud.

"We could discuss our plans and ideas with Google Cloud team members face to face, and they provided answers directly rather than pushing us

into a ticketing system," says Mackellar. "As well as the human interaction and customer support, we found the technology easier to use, more intuitive to set up and the services considerably more reliable than those of our previous provider."

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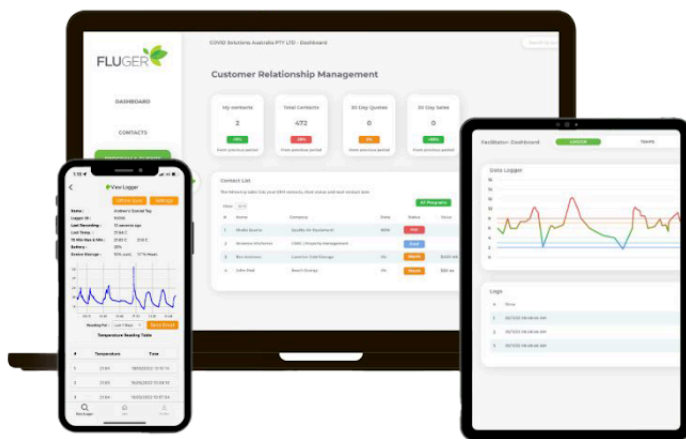
—Hamish Mackellar,
Founder, Flugger

Flugger began migrating to Google Cloud in early 2023, and completed the project six months later. The business is running Unix servers in Compute Engine with Apache PHP and JavaScript applications, and plans to use load balancers to autoscale its systems based on throughput. Its Cloud SQL (<https://cloud.google.com/sql>) instances are also running 12 different database systems across different geographies, helping the business retain data onshore to meet data sovereignty obligations in markets such as the United States and Australia.

Network services route data quickly and securely across the company's Google Cloud environment, while Cloud DNS (<https://cloud.google.com/dns>) provides a reliable domain name system, while protecting the business against malicious activity such as spoofing or denial of service attacks.

Cloud Storage (<https://cloud.google.com/storage>)

enables the business to store logs and other assets for its booking system in reliable, easily accessible storage buckets. In addition, a Google Maps Platform API and Locations API shorten the process of entering addresses into the booking system, and enables nurses and administrators to determine travel times.



Fluger dashboard across multiple devices

Turning to Google Cloud for a reliable, scalable and fast cloud infrastructure

With Google Cloud, Fluger has implemented a reliable, scalable and fast cloud infrastructure to power its workplace vaccination and health assessment booking software, while growing its pharmacy booking management system. "Our corporate health clients have noticed a huge improvement since we moved to Google Cloud," says Mackellar. "The speed with which our system

responds with a page, with the requested data, has more than doubled since Google Cloud, while infrastructure and system availability is close to 100 percent."

Fluger has also improved the speed and stability of its offerings cost-effectively. The company is reaping these rewards at a cost up to 15 percent lower than its previous cloud provider, which Mackellar describes as "fantastic value for money." This enables the business to direct more investment towards expansion and product development.

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The business anticipates robust growth from its pharmacy system that coordinates data workflows from initial booking through to updating government systems. This includes the Australian Immunisation Register, which records vaccines given to everyone in the country. "By managing and automating those flows, we can free up pharmacists' time and make their working days easier," says Mackellar.

The system also enables end-users to include multiple people into a single booking and book for multiple services. Each booking can accommodate different combinations or permutations of services, and the times needed for each. For example, a vaccination may take five minutes, a health assessment 25 minutes and a child immunization seven minutes, meaning a total required booking time of 37 minutes.

"With Fluger, an end-user can find time in a pharmacists' schedule, and reschedule easily if required," says MacKellar. As of September 2023, the business had about five pharmacies in Australia managing about 200 bookings per month with the system, and a similar-sized trial underway in the United States. "We now have the confidence to target a population of 330 million in the United States knowing that if a large pharmaceutical player wanted to roll us out to, say, 10,000 pharmacies, we're sorted out from an IT and cloud perspective," says MacKellar.



Hamish MacKellar, Founder, Fluger

Catching code issues before they impact performance

Google Cloud has enabled Fluger in identifying any problems with code written for updates, fixes and new functionality through SQL data analytics, before

it compromises the client or user experience. The business can then rewrite the code to eliminate bottlenecks before they cause a delay in service. In addition, Google Cloud's flexibility and ease of use means Fluger's lead developer and other team members can easily scale Compute Engine resources to accommodate the business's changing usage demands.

As Fluger grows, it plans to migrate to Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) to run its booking system in a dynamic containerized environment at scale, and explore the opportunities presented by the AI and machine learning technologies of Google Cloud. "There's a lot of room for AI, but it's a learning curve for us to understand and implement it into our system," says MacKellar.

"Overall, Google Cloud has been awesome," he concludes. "It combines the simplicity of setting up virtual machine instances quickly and easily with seamless integration and skilled service and support. We can maintain quality services and experiences for our clients and catch any problems early."

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